Contamination sweep: regime-dependent server behavior under declared attacks

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Contamination sweep (part 2)

This experiment compares server presets in one active-inference colony. Its legacy labels borrow FedGVI client-divergence vocabulary (Mildner et al. 2025), but the presets are not client losses or divergence theorems. `KLD` denotes the standard log-linear pool;

Contamination sweep (part 3)

every other label denotes a `robust_aggregate` heuristic constant in KLD (c=0.00), RKL (c=1.50), AR (c=1.30), beta (c=1.70), rcce (c=1.60). The estimand is consensus mass $q(\text{true state})$ under the declared contamination mechanism.

Contamination sweep (part 4)

The colony contains 7 sentinels, of which 2 mix their broadcasts toward a confident-wrong delta at each contamination rate. The comparisons remain conditional on this fixed world, attack target, and preset grid.

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Contamination sweep (part 5)

The evidence map in fig. 9 locates these comparisons in the conditional heuristic-server lane. It keeps the fixed-world matched trials distinct from the seed-level review below and prevents either result from inheriting client-loss or variational-server guarantees.

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Contamination sweep (part 6)

As the contamination rate rises across $\{0, 0.225, 0.45, 0.675, 0.9\}$, the **standard** (KLD) consensus accuracy degrades monotonically:

Contamination sweep (part 7)

Contamination rate	KLD	RKL	AR	beta	rcce
0	1.000	0.997	0.998	0.996	0.996
0.225	0.999	0.993	0.995	0.990	0.992
0.45	0.995	0.987	0.990	0.984	0.986
0.675	0.975	0.985	0.989	0.981	0.983

Contamination sweep (part 8)

As the saboteurs capture more belief mass, `KLD` falls monotonically while at least one non-reference preset remains above the stated accuracy threshold. **Reference trend:** standard accuracy degrades monotonically with rate, recorded as Yes.

Contamination sweep (part 9)

Worst-rate display check: at rate 0.900, at least one non-reference preset remains at or above 0.50, recorded as Yes. Standard accuracy is 0.6928; the highest robust consensus mass in this single-world mechanistic sweep is 0.9880, attained by AR.

Contamination sweep (part 10)

The rate trend above is one deterministic sweep per cell. The paired profile reruns each rate over $n_{\text{trial}} = 960$ matched trials nested within the fixed seeded world. Each non-reference preset is compared with the standard pool, and p-values are BH-adjusted within method (Benjamini and Hochberg 1995).

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Contamination sweep (part 11)

This is a finite, rate-resolved diagnostic, not a continuous-family result.

Contamination sweep (part 12)

Server preset	Rate	Rank-biserial-derived d-equivalent	Label
RKL	0	saturated (r=-1)	large

Contamination sweep (part 13)

Server preset	Rate	Raw p	q	Reject
RKL	0	1.11e-158	1.85e-158	Yes
RKL	0.225	1.11e-158	1.85e-158	Yes
RKL	0.45	1.11e-158	1.85e-158	Yes
RKL	0.675	1.88e-155	2.35e-155	Yes

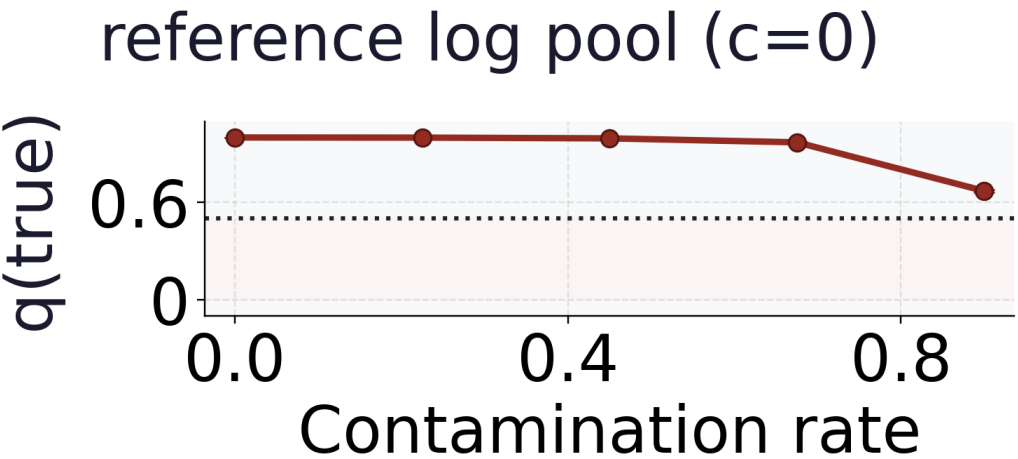
Contamination sweep (part 14)

The KLD baseline is excluded from both projections because it is the standard reference, not a self-contrast. The displayed d-equivalent is a rank-biserial-derived transform, not raw Cohen's d; the signed-saturation marker flags contrasts where the rank-biserial correlation saturates at ± 1.

Contamination sweep (part 15)

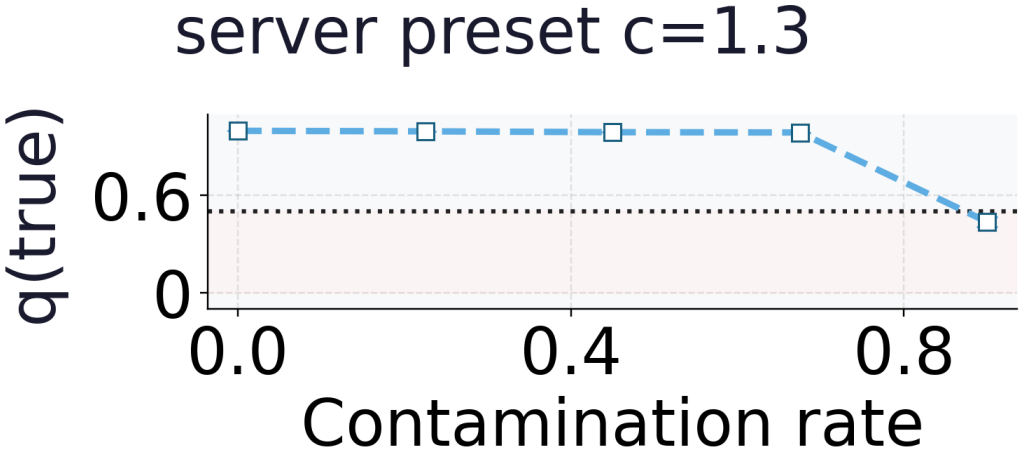
These contrasts decorate the server-side `robust_aggregate` heuristic only.

Contamination sweep (part 16)



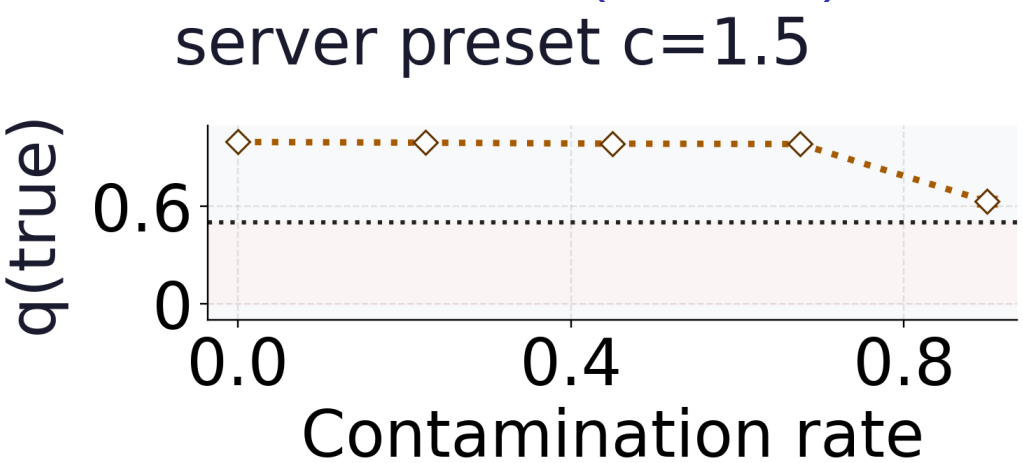
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Contamination sweep (part 17)



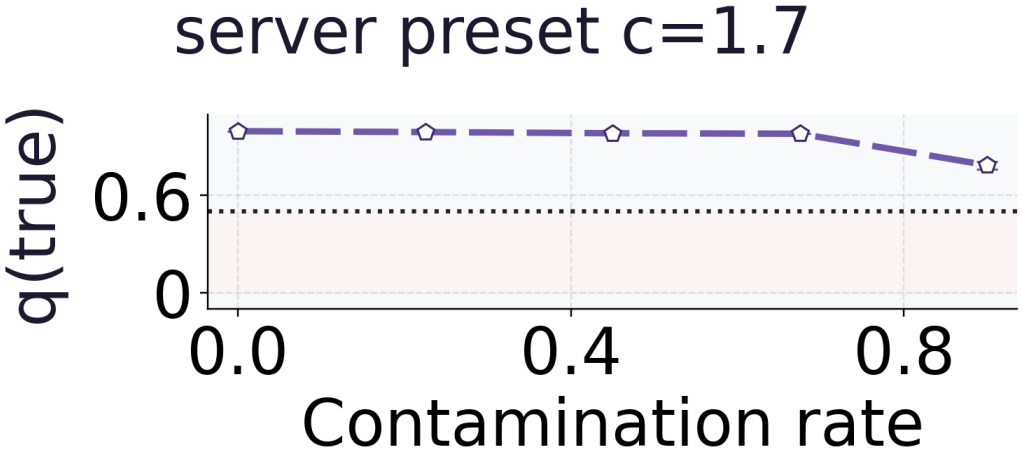
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Contamination sweep (part 18)



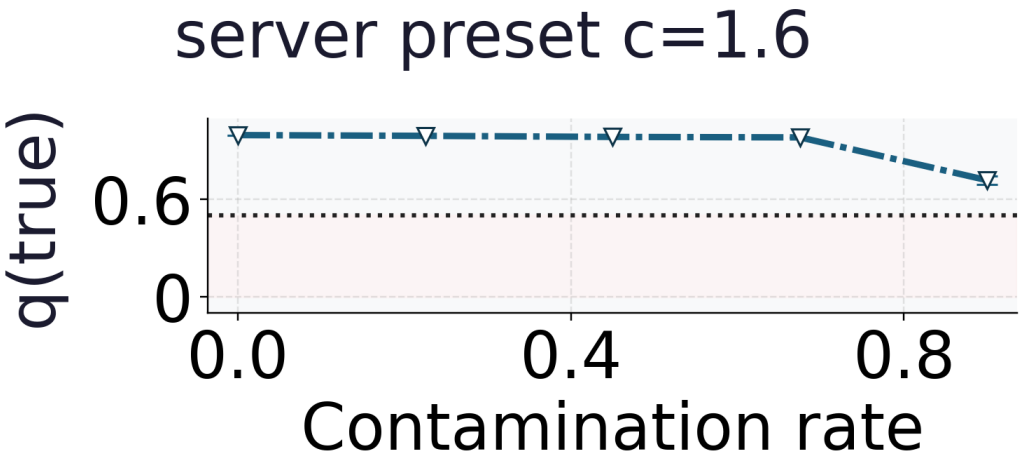
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Contamination sweep (part 19)



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Contamination sweep (part 20)



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Contamination sweep (part 21)

960 matched trials per rate. Whiskers: each preset's own percentile-bootstrap interval.

Contamination sweep (part 22)

Predeclared floor: 0.5. The pale band lies below this reference.

Contamination sweep (part 23)

Largest max-rate preset minus reference: +0.116. Max rate: 0.9.

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Contamination sweep (part 24)

Consensus accuracy here means q(true state). These configured server presets are not client-loss implementations or a complete source protocol.

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Selection-free all-method robustness review

The comparison in fig. 17 is the principal comparative robustness surface.

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Selection-free all-method... (part 2)

It joins the conditional-world cells to the directional rate profiles while retaining every configured non-reference server preset. It uses 160 deterministic seed replicates and 24 trials nested within each seed and cell.

Selection-free all-method… (part 3)

Read the portrait surface from top to bottom: Panel A is the full-width conditional heatmap, and Panels B–D stack the three rate profiles. Their elbow leaders terminate in reserved right-side label lanes so curve identity remains separate from the plotted endpoints.

Selection-free all-method... (part 4)

The finite attack union is clean, confident wrong, permutation, byzantine, drift, label noise, uniform. The rate-resolved directional mechanisms are confident wrong, byzantine, drift, with entropy controls uniform, label noise and registered rates $\{0, 0.2, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$.

Selection-free all-method... (part 5)

The independent unit is seed within a declared scenario or rate cell; the nesting rule is n_trials nested within each seed/cell; no trial is promoted to an independent world.

Selection-free all-method... (part 6)

The payload is selection-free source payload; every configured non-KLD method is reported at every directional rate and no winner is used for inference, and its statistics surface is selection-free. It retains seed-level contrasts, paired Wilcoxon/rank-biserial results, percentile-bootstrap intervals, MCSE,

Selection-free all-method... (part 7)

an observed-effect MDE, and BH-adjusted rate families. BH ownership is BH is applied within each attack-mechanism $\times$ method rate family; cells sharing design structure are not treated as independent families.

Selection-free all-method... (part 8)

The precision plan targets maximum MCSE 0.0100; the observed maximum is 0.0066 across 96. No method is selected by seed, rate, or pooled mean. Mixed and reversed cells remain visible, and the grid does not close the open calibration, external-data, or server-theory questions.

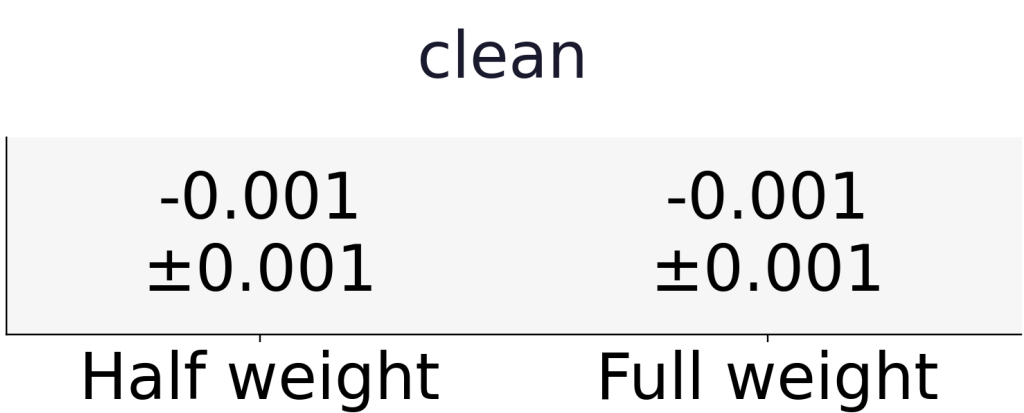
Selection-free all-method... (part 9)

The source-generated exact-value fallback calls the same finite grouping helper as Panel A. For every displayed attack-by-weight cell it records the printed grouped mean and the printed half min–max span, preventing the numeric fallback from drifting from the renderer.

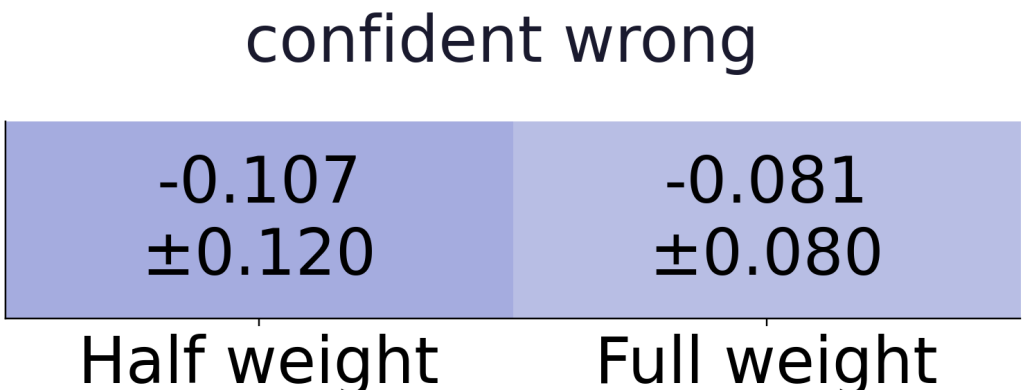
Selection-free all-method… (part 10)

Conditional preset-minus-reference means
± half min–max span across declared cells; not a confidence interval.

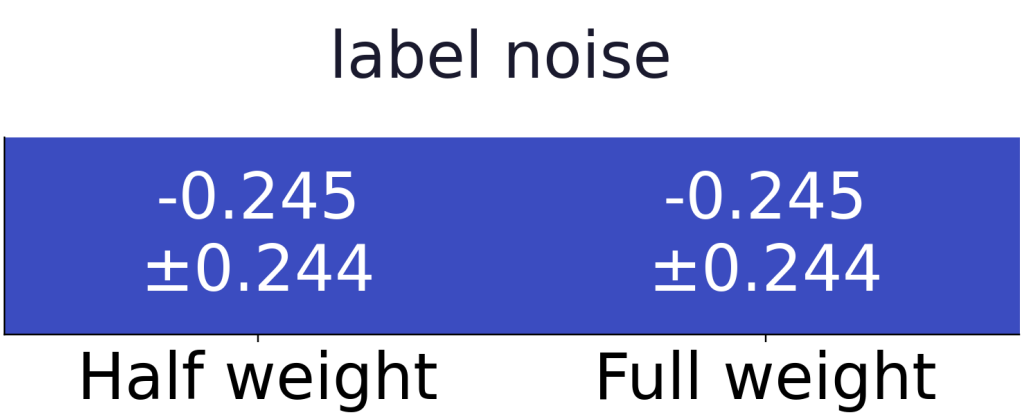
Selection-free all-method… (part 11)



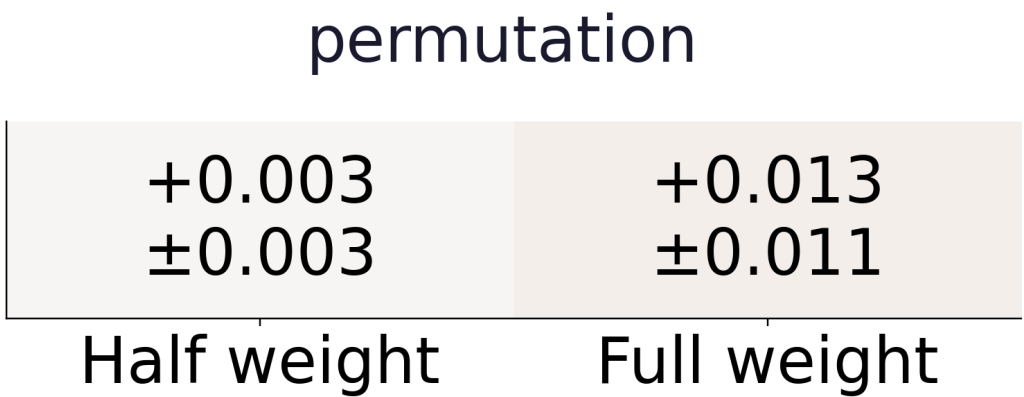
Selection-free all-method… (part 12)



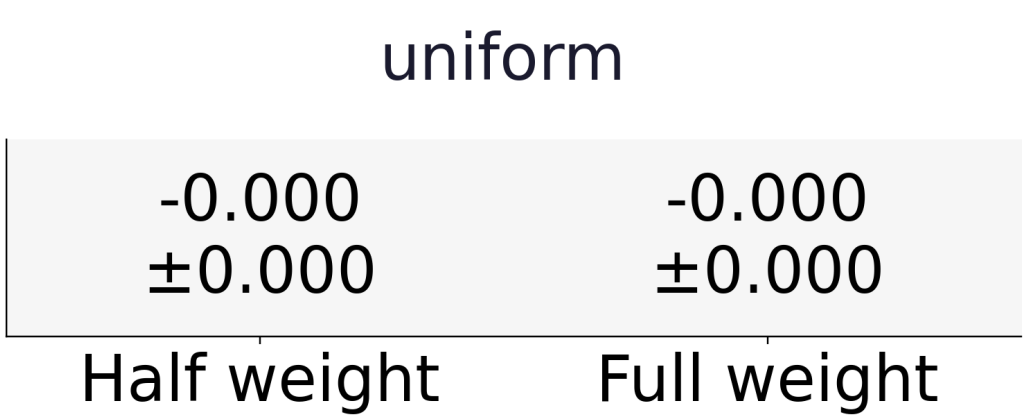
Selection-free all-method... (part 13)



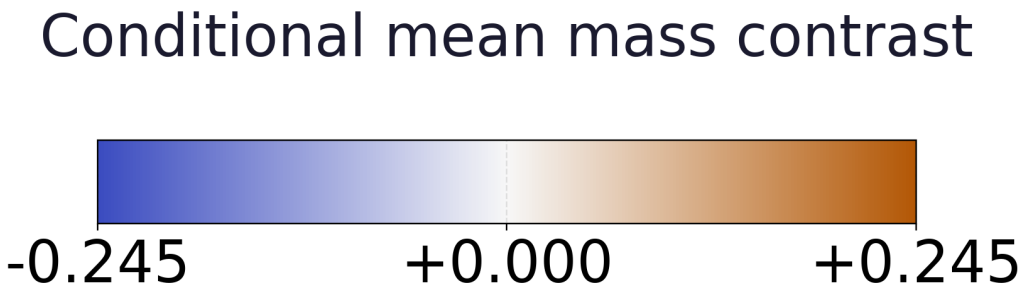
Selection-free all-method… (part 14)



Selection-free all-method… (part 15)



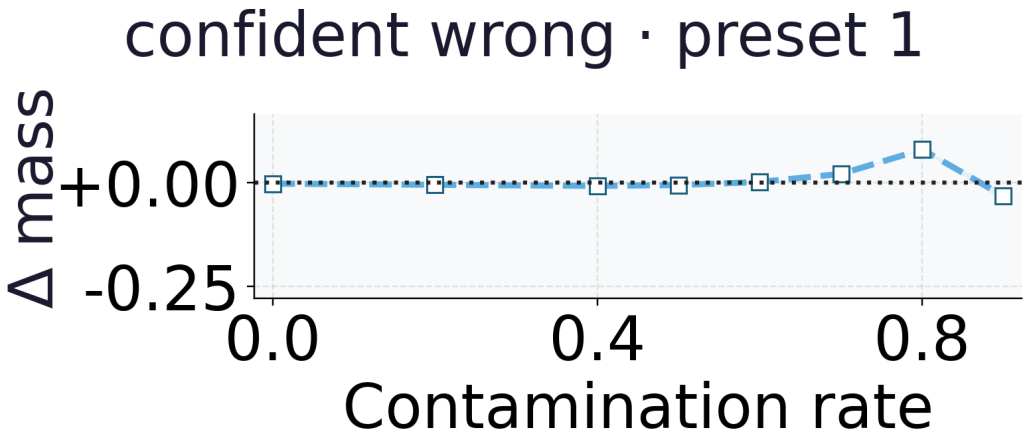
Selection-free all-method… (part 16)



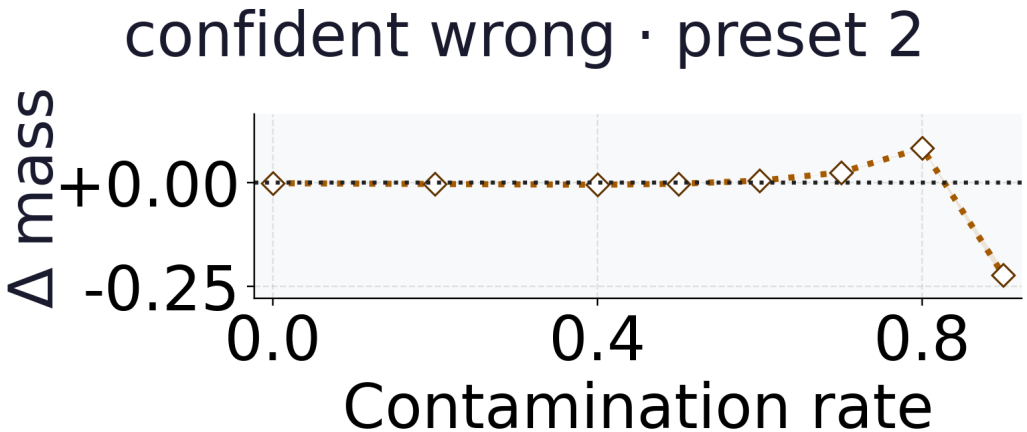
Selection-free all-method… (part 17)

Each preset retains its own 95% seed-bootstrap band.
Y limits are shared within each mechanism. All predeclared presets are shown.

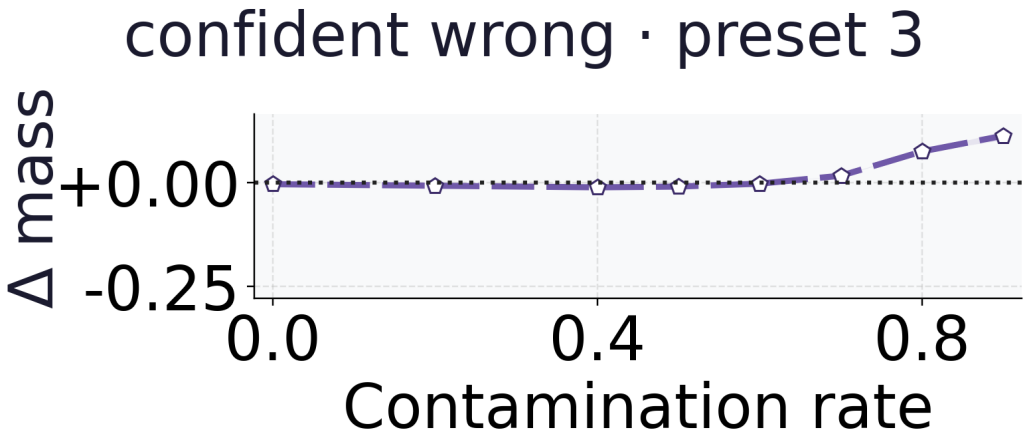
Selection-free all-method… (part 18)



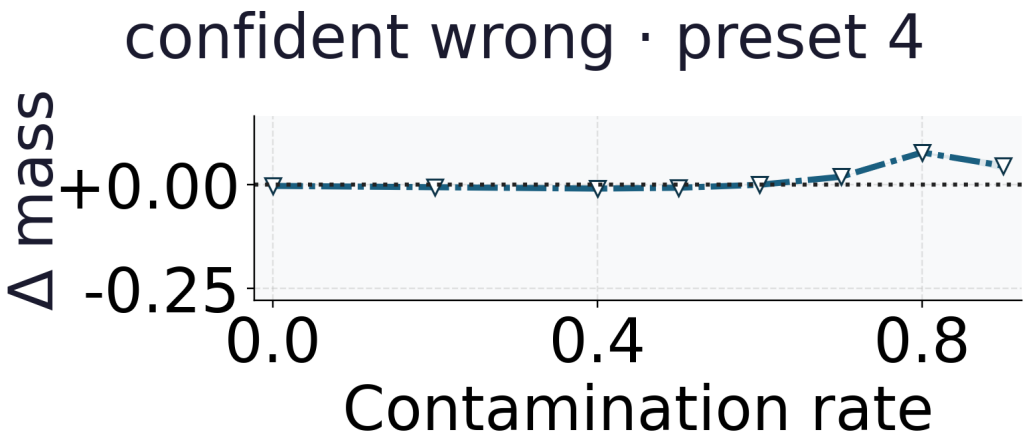
Selection-free all-method… (part 19)



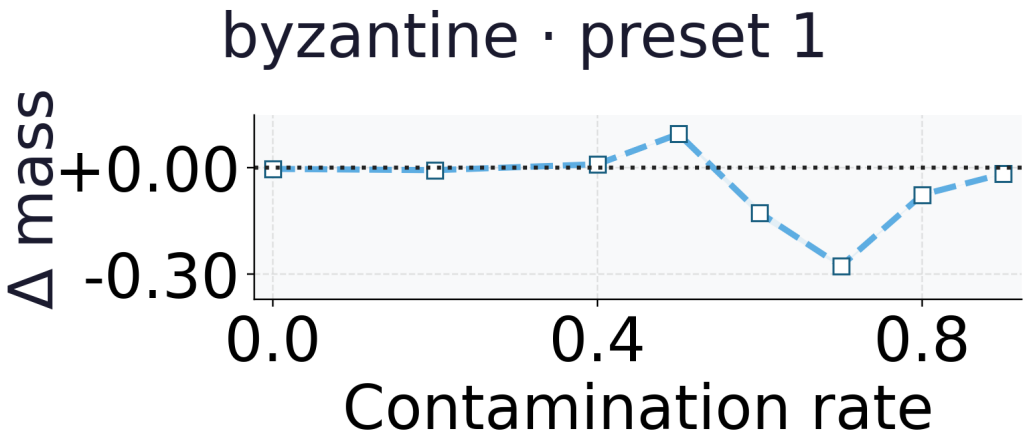
Selection-free all-method… (part 20)



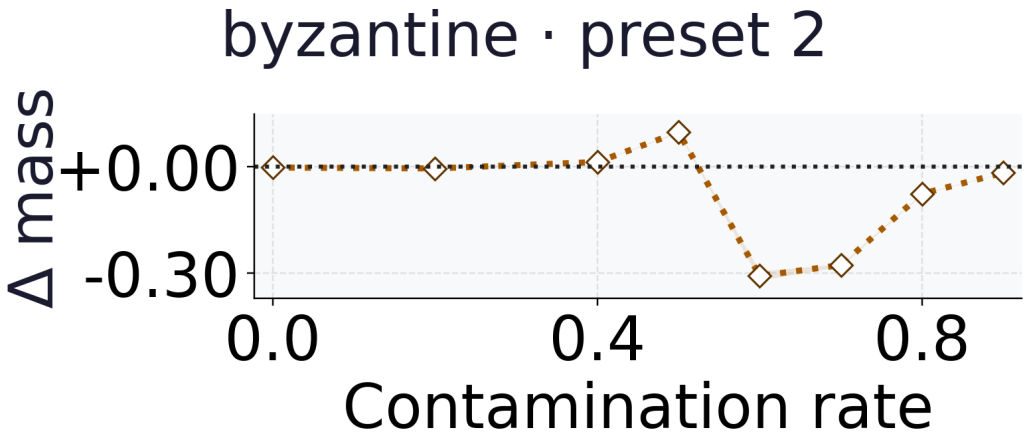
Selection-free all-method… (part 21)



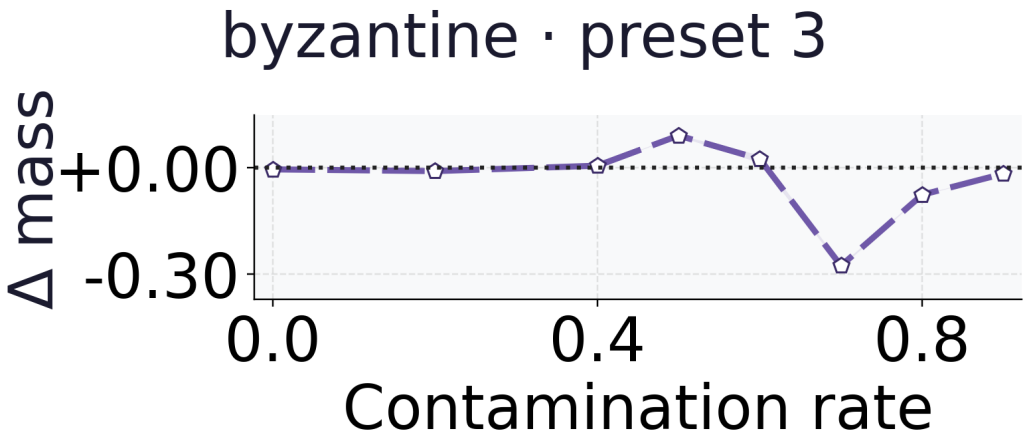
Selection-free all-method… (part 22)



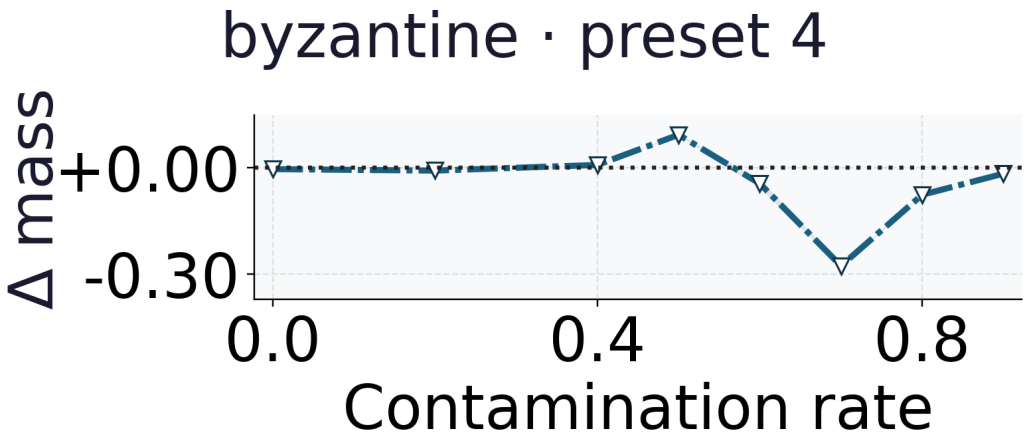
Selection-free all-method… (part 23)



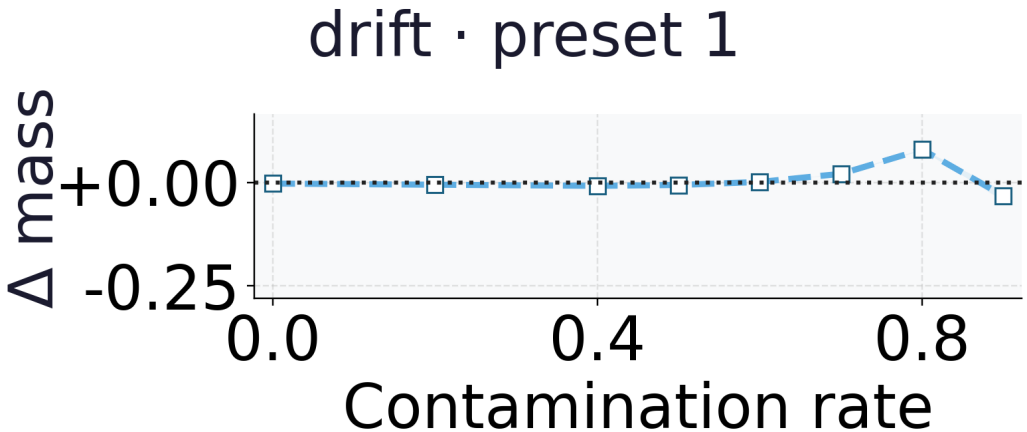
Selection-free all-method… (part 24)



Selection-free all-method… (part 25)

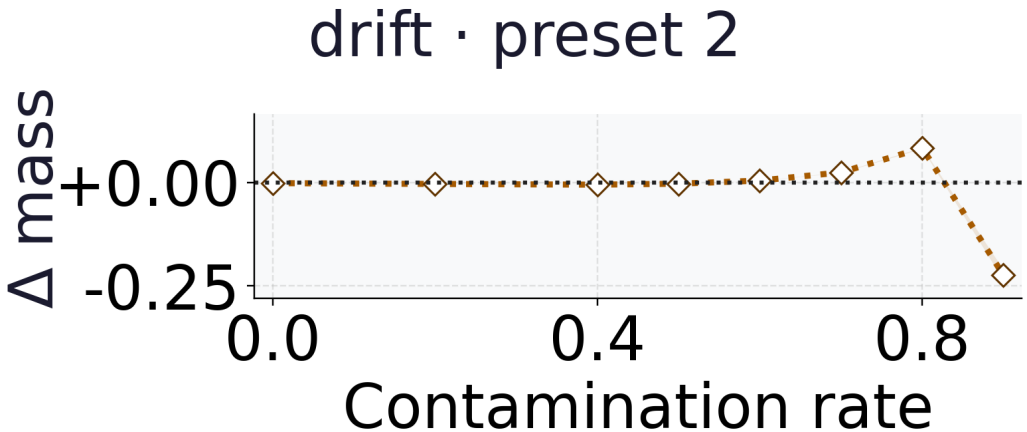


Selection-free all-method… (part 26)



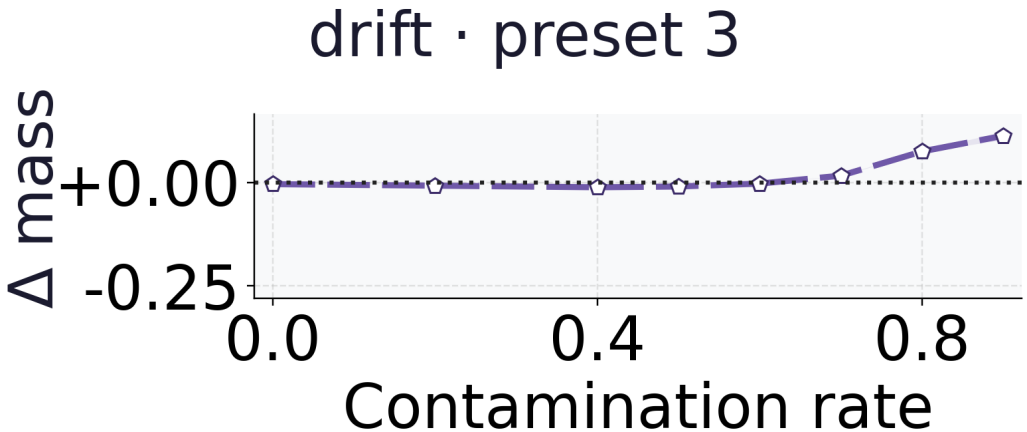
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Selection-free all-method… (part 27)



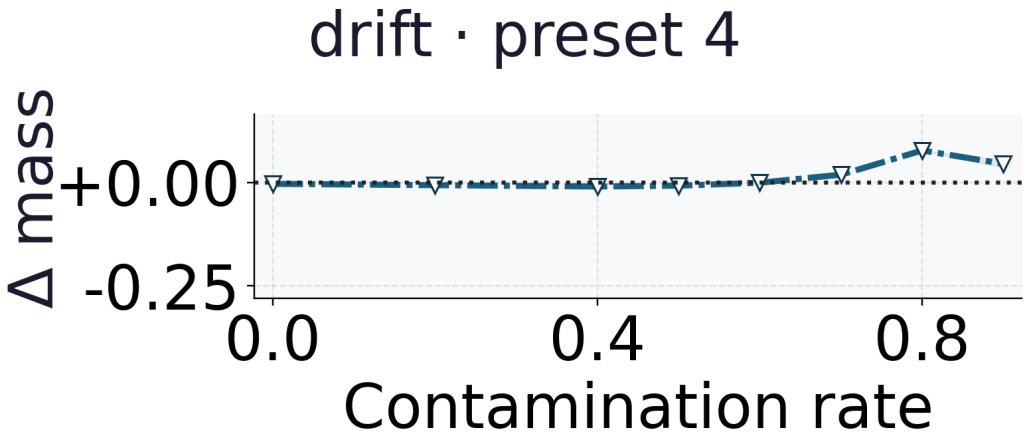
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Selection-free all-method… (part 28)



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Selection-free all-method... (part 29)



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Conditional matched-trial comparison at the declared rate

At contamination rate 0.800, each of 960 matched trials redraws the heterogeneous healthy colony while holding the seeded world's true state and attack target fixed.

Conditional matched-trial… (part 2)

Each non-reference server preset is compared with the standard pool using the paired Wilcoxon signed-rank test (Wilcoxon 1945).

Benjamini–Hochberg FDR adjustment applies within the declared family (Benjamini and Hochberg 1995).

Conditional matched-trial... (part 3)

The generated `Wins` field is true only when the adjusted null is rejected and the paired effect is positive. It is conditional bookkeeping for this fixed world and rate, not a universal method ranking.

Conditional matched-trial… (part 4)

Server preset	Effect size	Raw p	q	Wins
AR	1.0000	1.11e-158	1.11e-158	Yes
RKL	1.0000	1.11e-158	1.11e-158	Yes
beta	1.0000	1.11e-158	1.11e-158	Yes
rcce	1.0000	1.11e-158	1.11e-158	Yes

Conditional matched-trial… (part 5)

Method	Derived d_{eq}	Label	Mean Δ accuracy	95% CI
AR	saturated (r=+1)	large	0.0846	[0.0821, 0.0873]
RKL	saturated (r=+1)	large	0.0809	[0.0783, 0.0834]

Conditional matched-trial… (part 6)

Method	Raw p	q	Power	Target n_{trial}	Reject
AR	1.11e-158	1.11e-158	1.0000	5	Yes
RKL	1.11e-158	1.11e-158	1.0000	5	Yes

Conditional matched-trial… (part 7)

Method	n	Mean acc. @ verdict rate	95% CI
KLD	960	0.9021	[0.8993, 0.9049]
RKL	960	0.9829	[0.9827, 0.9832]

Conditional matched-trial… (part 8)

Reference mean. Naive-pool mean accuracy at the verdict rate is 0.9021 (per-trial mean over 960 trials; the bootstrap CI is in tbl. 10).

Positive contrasts. At least one non-reference preset is a BH-rejected positive contrast: Yes.

Conditional matched-trial… (part 9)

Headline display. The method is RKL (tied set: RKL, AR, beta, rcce), with rank-biserial-derived d-equivalent saturated (r=+1) (large), mean accuracy difference 0.0809 ($[0.0783, 0.0834]$), raw $p = 1.11 \times 10^{-158}$, and $q = 1.11 \times 10^{-158}$.

Conditional matched-trial... (part 10)

Its observed-effect design power is 1.0000.
Prospective n for power 0.80 is 5.

Conditional matched-trial… (part 11)

Display rule. Under largest positive rank-biserial effect_size; stable method order tie-break (tie-break: first robust method in divergences order), observed-effect design power is 1.0000 at the run's $n_{\text{trial}} = 960$.

Conditional matched-trial... (part 12)

A confirmatory replication should budget $n_{\text{trial}} = 5$ for power 0.80 (prospective $n = 7$).

Conditional matched-trial... (part 13)

The standard pool degrades and some non-reference presets have positive conditional contrasts. The statistics module produces the paired tests, multiple-testing adjustment, bootstrap intervals, and planning analysis; the prose does not promote them beyond their matched-trial estimand.

Conditional matched-trial... (part 14)

The headline label is a deterministic display choice, not a unique scientific winner. The complete tied set is RKL, AR, beta, rcce, the largest paired mean-difference method is AR, and the worst-rate pooled display method is beta. These may differ because they answer different descriptive questions.

Server-side heuristic axis: conditional contrasts without theorem transfer

The comparison above belongs to the heuristic server axis defined in sec. 6.3. fig. 18 visualizes the `robust_aggregate` divergence-reweighting that down-weights agents at pooling time.

Server-side heuristic axis (part 2)

Its positive formal property is the naive-recovery limit of Theorem 5 (eq. 10, the 0 residual of sec. 10.1),

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Server-side heuristic axis (part 3)

and it has a scoped no-go result for a declared separable objective class, not an objective certificate.

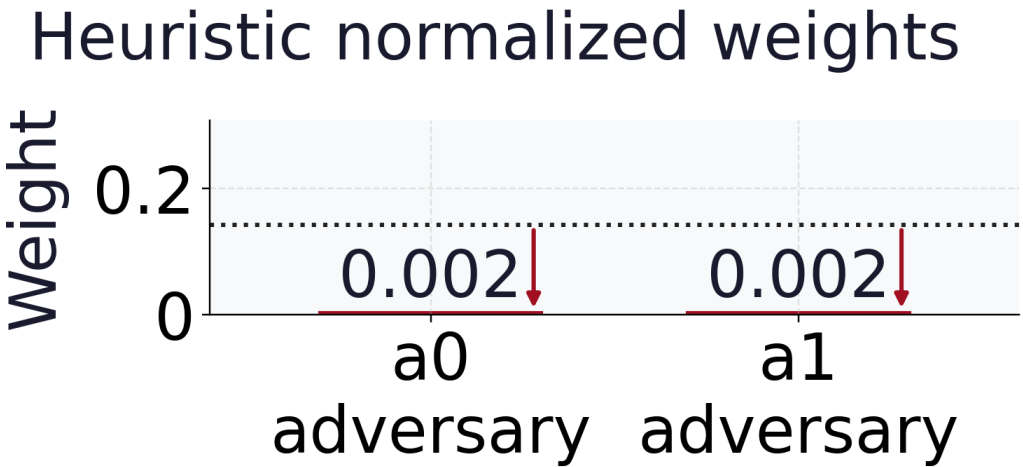
Server-side heuristic axis (part 4)

The heuristic carries no bounded-influence guarantee. The effect sizes, intervals, and planning quantities characterize its conditional contrasts; they do not certify the per-agent generalized-Bayes claim, which remains separately source-conditional in sec. 10.6.

Server-side heuristic axis (part 5)

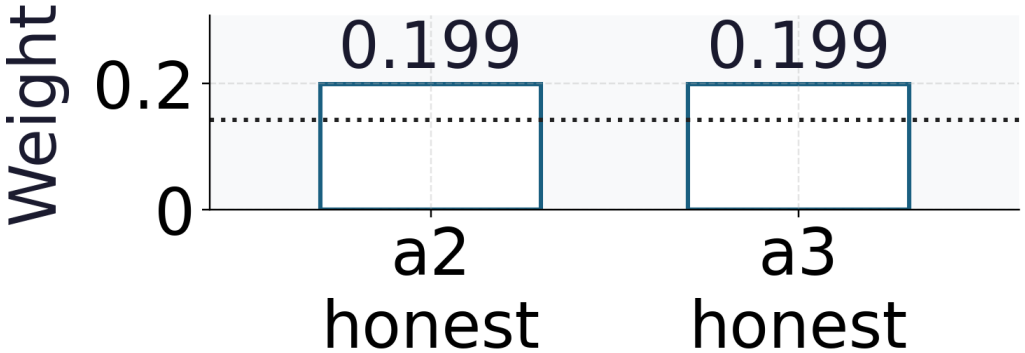
Open fill: honest. Cross-hatch: adversary.
Dotted line: equal-weight pool.
All panels share the same weight scale.

Server-side heuristic axis (part 6)



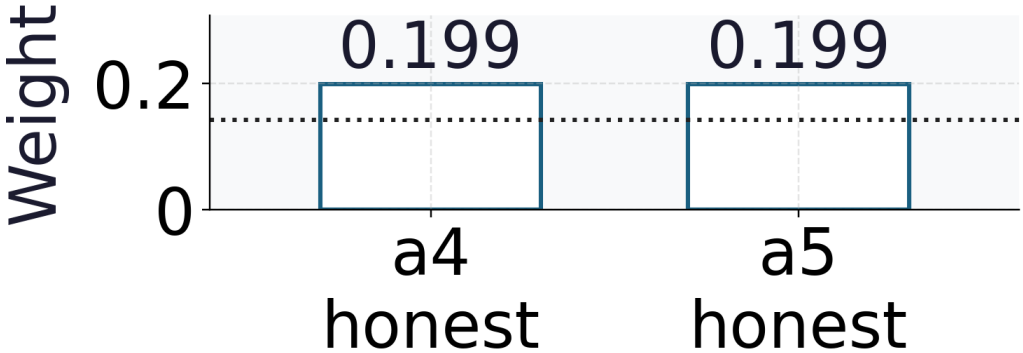
Server-side heuristic axis (part 7)

Heuristic normalized weights



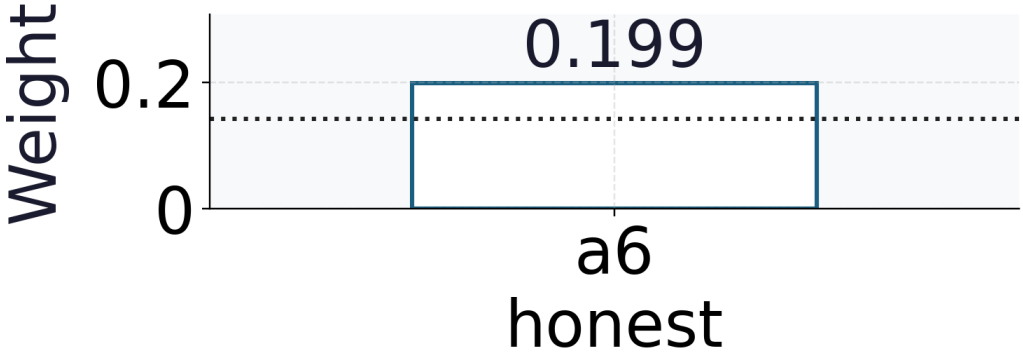
Server-side heuristic axis (part 8)

Heuristic normalized weights



Server-side heuristic axis (part 9)

Heuristic normalized weights



Server-side heuristic axis (part 10)

Agents contaminated: 2/7 (29%)
Weight sum: 1.000
Healthy mean weight: 0.199

Server-side heuristic axis (part 11)

Adversary mean weight: 0.002
Observed role-weight gap: 0.198

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Server-side heuristic axis (part 12)

Server heuristic weights support a recovery-limit diagnostic. They do not inherit client-loss robustness guarantees.

Variational aggregator: conservative objective-backed weight control

The heuristic has the higher point accuracy in the configured verdict cell. The variational aggregator of sec. 8.8.2, derived in sec. 15, is its objective-backed complement.

Variational aggregator (part 2)

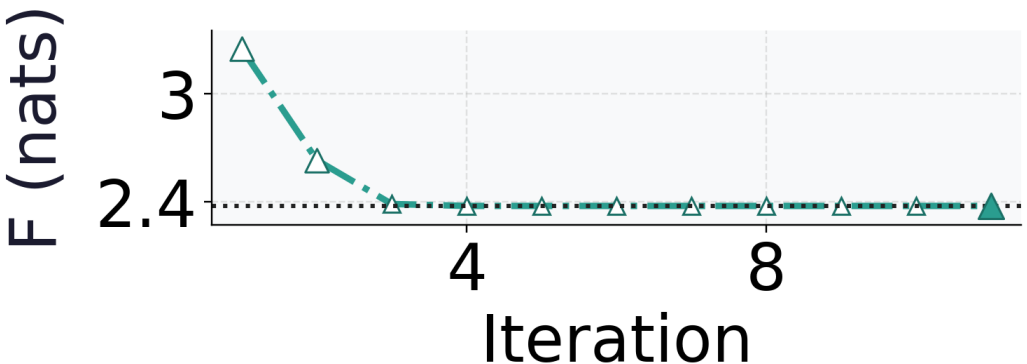
Each exact block update does not increase the stated free energy eq. 12, and a converged fixed point is coordinatewise stationary. Two deterministic runs show its weight behavior at robustness 1.50.

Variational aggregator (part 3)

First, the descent. On a contaminated colony the free energy falls monotonically from 3.2458 to 2.3780 (a drop of 0.8678 over 11 block-coordinate iterations, converged: Yes), with a largest single-step *increase* of 8.88e-16 — machine zero, the numerical witness of the descent theorem (sec. 15.3).

Variational aggregator (part 4)

Variational objective descent



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Variational aggregator (part 5)

Triangles and dash-dot: variational objective.
Filled triangle: final iterate.
Large open triangles: largest observed step.

Variational aggregator (part 6)

Iterations: 11
Total drop $\Delta F = 0.868$ nats
Max ascent step = 8.9e-16 nats

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Variational aggregator (part 7)

Monotone: yes ✓ converged

Variational aggregator (part 8)

One deterministic trajectory; no sampling interval.
A final iterate is not a global-optimum certificate.

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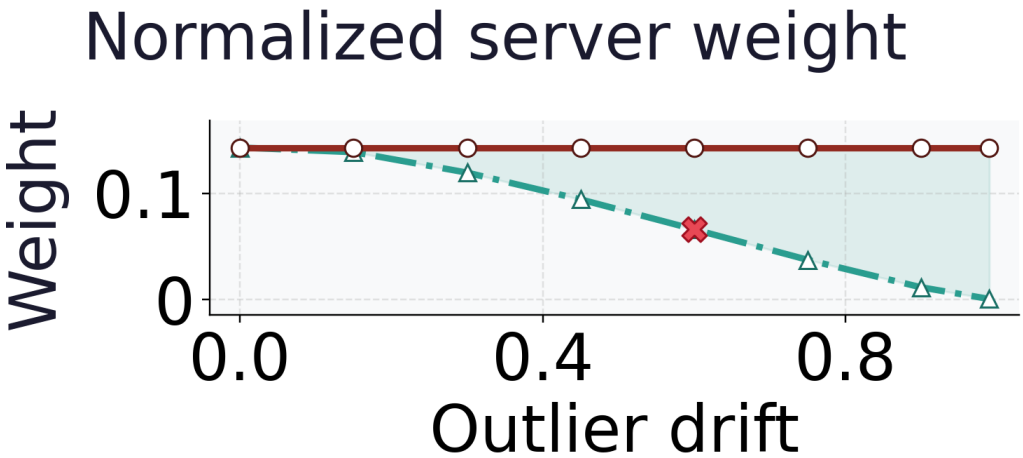
Variational aggregator (part 9)

Second, the effective-weight response. As one agent is drifted from healthy toward a confident-wrong delta, its normalized influence falls from 0.143 to below 0.001 — a factor of 267.1 below the fixed 0.143 the naive log-linear pool grants every agent regardless of how wrong it is.

Variational aggregator (part 10)

The gap between the falling variational curve and the flat naive line is the empirical redescending weight response, drawn.

Variational aggregator (part 11)



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Variational aggregator (part 12)

Triangle/dash-dot: variational.
Circle/solid: naive fixed weight.
Shading: deterministic weight gap.

Variational aggregator (part 13)

Outlier drift: 0 is consensus; 1 is confidently wrong.
This is a configured path, not estimator-level B-robustness.

Variational aggregator (part 14)

Variational final weight: 0.000535
Naive fixed weight: 1/n = 0.143

Variational aggregator (part 15)

Cross: first configured drift below 0.5 × naive. Drift = 0.60

Variational aggregator (part 16)

The honest trade is conservatism: because F carries the $-H(q)$ entropy term, its consensus is the maximum-entropy distribution consistent with the weighted cross-entropies, deliberately flatter than the product-of-experts.

Variational aggregator (part 17)

The variational aggregator therefore does *not* win the peak-accuracy verdict of sec. 10.5.2 — that remains the sharp heuristic's role — and the two are reported as complements, never conflated: rigor-with-conservatism on one side, accuracy-without-an-objective on the other.